

Video wall: status and next steps

Old Samsung monitors as one screen

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Where the video wall stands on 29 September, the plan to the finished wall, and the bench test for tomorrow.

Bottom line

Ready for a bench test, not for the wall

Tomorrow can start with step 0, the bench pilot with two screens, A1 and B1. The setup script is written and tested only on a fake card; it has never run on a real Pi. No wall software (server or Pi client) exists yet, so tomorrow is hardware and setup, not synced playback.

What's done

Plan and design	README + ROADMAP in the videowall repo	Done
Pi setup script	scripts/pi-setup.sh, parallel over SSH, safe to rerun	Written
Boot-file test	tests/test_pi_setup.sh, fake card	Passes
Script on a real Pi	first run tomorrow	Not yet
Lab access	Tailscale; node techlab-01 is jump host, resolves wall-a1.local	Ready
TV Pi experience	Pi 3 B kiosk live; same OS and tuning path	Done
Several screens same page in sync	TV pages follow the clock	Done
Render server + Pi client	roadmap steps 1-3	Not started
Office control (wall_state / wall_status)	step 5, last	Not started

Hardware

Item	State
Monitors	Samsung SyncMaster 720N, 17 in, 1280x1024, VGA only; 5x3 = 15 now, then 5x5, target 6x6
Computers	Raspberry Pi 3B+, one per screen, wired Ethernet
Adapters	HDMI to VGA; 2-3 models to compare on the bench, then buy one model for all
Switch	TP-Link TL-SG1024D, 24 ports: enough for 15 + server, not for 25
Power	Shared USB charger, 5 V / 2 A per port, below the 2.5 A a Pi 3B+ wants: undervoltage risk
Mount	3D-printed bracket, measurements to redo before printing

Overall plan

1. Step 0, Pis and bench: flash all cards, run the setup script, pick the adapter, settle Pi 3 video memory and hardware decode.
2. Step 1, Mosaic with stills: pool folder on the node share, each Pi cycles its images, test pattern, Identify Screen. 5x3.
3. Step 2, Videowall with stills: FFmpeg builds one canvas (6400x3072), slices it, synced image change.
4. Step 3, Video: tile render, preload, play_at, chrony; measure drift across 15 screens.
5. Step 4, Composition: Fit/Fill/Center, logo, title, credits, frame.
6. Step 5, Office link: wall_state and wall_status tables, Video wall screen in the office.
7. Step 6, Physical build: bracket, power fix, second switch, 5x5 then 6x6.

Steps for tomorrow

1. Bring: two Pi 3B+ with cards, two 720N monitors side by side (A1 left, B1 right), 2-3 HDMI-VGA adapter models, a proper 2.5 A supply and the shared USB charger, Ethernet cable.
2. Flash both cards on the Mac with Raspberry Pi Imager: Raspberry Pi OS Lite (64-bit), hostnames wall-a1 and wall-b1 (grid code: letter = column, number = row, A1 top-left, E5 bottom-right of a 5x5), user techlab, SSH public-key only with ~/.ssh/techlab.pub, time zone Europe/Lisbon, Wi-Fi = lab network with country PT (2.4 or 5 GHz).
3. Before ejecting: scripts/pi-setup.sh --boot /Volumes/bootfs (sets 1280x1024 output on the card).
4. Boot on the lab network, check: ping wall-a1.local, ping wall-b1.local.
5. Run: scripts/pi-setup.sh wall-a1.local wall-b1.local, then reboot if it says REBOOT NEEDED. Read logs/wall-a1.local.log and logs/wall-b1.local.log.
6. Check power: throttled=0x0 in the report, on both the 2.5 A supply and the shared charger.
7. Try each adapter: picture at 1280x1024 at 60 Hz, sharp, no black border.
8. Play the test tile (~Downloads/wall-test/tile-1280x1024.mp4, command in videowall docs/pi-setup.md) with mpv; watch for dropped frames. Note what video memory setting it needs.
9. Write the results in videowall docs/pi-setup.md and ROADMAP.md; if the pilot is good, flash the rest of the grid (5x3: wall-a1 to wall-e3).

Open questions

Decide soon

Which machine hosts the wall server: the edge node (has nginx, Samba, FFmpeg) if its CPU copes, or an old laptop. Power: buy 2.5 A supplies or a stronger charger. Repo public or private.